

[[Lab Title]]: [[Accent Word]] [[rest of title]]

[[U0 / S0.3]]

NAME

DATE

PERIOD

GROUP MEMBERS

GOALS

- [[Operate a specific tool correctly — name the tool and the skill.]]
- [[Measure a specific quantity and explain why the wrong tool won't do.]]
- [[Determine / calculate the target quantity and state what you do with it.]]
- [[Build or set up the apparatus and accomplish the physical task.]]
- [[Record data in your lab notebook as you take it — not from memory afterward.]]

PURPOSE

[[You are going to ____ using nothing but _____. Both jobs depend entirely on whether you can use the tools correctly. That is the real point of today.]]

BACKGROUND

[[**Concept 1 lead-in.**]] [[Two to four sentences. Include the specific misstep students make — that sentence is what makes this section worth printing.]]

[[**Concept 2 lead-in.**]] [[Explanation.]]

[[NAME OF THE RELATIONSHIP]]

[[quantity = quantity ÷ quantity]] · [[units]]

[[One sentence on why the relationship behaves the way it does — the sentence that connects the math to the identification or conclusion.]]

[[REACTION / EQUATION FOR TODAY]]

[[$\text{CaCl}_2(\text{aq}) + \text{Na}_2\text{CO}_3(\text{aq}) \rightarrow \text{CaCO}_3(\text{s}) + 2 \text{NaCl}(\text{aq})$]]

PERCENT ERROR

percent error = $(| \text{experimental} - \text{accepted} | \div \text{accepted}) \times 100$

SAFETY

- [[Splash goggles go on before any bottle is opened and stay on until every piece of glassware is washed and put away — including while other groups are still working.]]
- [[Chemical + concentration]] is [[hazard]]. On skin: [[exact response and duration]]. In eyes: [[eyewash duration, hold eyelids open, send a group member for me immediately]]. Do not wait to see if it gets better.
- [[Equipment-specific hazard and the action that prevents it.]]
- Check the rim of every [[piece of glassware]] before you use it. Chipped glass cuts. Broken glass goes in the glass box by the back sink — never the trash can.
- [[Thermal / electrical / burner statement, or: "Nothing is heated today and no burner is lit."]]
- Report every spill, including water. Water on the floor is how people fall.

PRE-LAB — DUE AT THE START OF THE PERIOD

Answer all of these in your lab notebook, numbered 1 through [[N]], before you walk in. Do not write on this handout — it stays clean so you can follow the procedure with wet hands. You will not be allowed to start the lab without the pre-lab finished, because most of these are the parts people get wrong when they wing it.

- 1 [[CONCEPTUAL — a distinction they will otherwise blur. Name the word you want used.]]
- 2 [[CONCEPTUAL — why one tool/method is trusted over another.]]
- 3 [[CONCEPTUAL — what has to be true for the observed phenomenon to happen, in their words.]]
- 4 [[PROCEDURAL — Read Part __. Exactly what quantity do you measure, and with what tool?]]
- 5 [[PROCEDURAL — In Part __, step __, where does __ end up, and what problem does that solve?]]
- 6 [[PROCEDURAL — Which part do you start first, and why does that order matter for finishing on time?]]
- 7 [[PROCEDURAL — What do you do with __ when you are finished with it?]]
- 8 [[SAFETY — Specific exposure scenario. State exactly what you do, and write where the eyewash station / extinguisher / shutoff is in this room.]]
- 9 Set up your [[N]] data tables in your notebook. Copy them exactly as shown, in ink, leaving enough room to write in each blank. Units go in the row labels — do not plan to add them later.

TABLE 1 · [[TABLE NAME]]

[[Row label with (units)]]
[[Row label with (units)]]
[[Row label with (units)]]
[[Row label with (units)]]
[[Row label with (units)]]

TABLE 2 · [[TABLE NAME]]

[[Row label]]
[[Row label with (units)]]
[[Row label with (units)]]
[[Row label with (units)]]
[[Row label with (units)]]

TABLE 3 · [[TABLE NAME]]

[[Qualitative observation row]]
[[Row label]]
[[Row label with (units)]]
[[Row label with (units)]]
[[Row label with (units)]]

[[next class]]

If you write a number wrong during the lab, draw one line through it and write the correct value beside it. Never scribble it out and never erase — I need to see what you originally recorded.

MATERIALS

Per group: [[item]] · [[item with size]] · [[item (2)]] · [[item]] · [[0.5 M reagent]] · [[0.5 M reagent]]

PROCEDURE**Part A — [[Part Name]]**

1. [[Major step.]]
 - [[Substep — the check or condition.]]
 - [[Substep — what to do if the check fails.]]
2. [[Major step with the exact quantity and precision: "Record its mass to 0.01 g."]]
3. [[Major step.]]
4. [[Major step.]]
 - [[The thing they must write down before moving on.]]

Part B — [[Part Name]]

1. [[Major step.]]
2. [[Major step with reading precision: "Read at eye level and record to 0.1 mL."]]
3. [[The step that tells them not to "fix" their data: "Do not go back and adjust — the disagreement is the data."]]
4. [[Immediate cleanup step.]]

Part C — [[Part Name]] [[START THIS BEFORE PART D]]

1. [[Measure exact volume with the named tool and transfer to the named vessel.]]
2. [[Rinse / reset step between reagents.]]
3. [[The step where the thing happens.]]
 - Watch the instant [[they meet]]. Write what you see in your notebook before you do anything else — [[name the three properties: color, clarity, whether it settles]]. "It changed" is not an observation.
4. [[Build the apparatus:]]
 - [[Substep with a real dimension: "about 15 cm above the base."]]
 - [[Substep.]]
 - [[Substep + the reason it matters: "This stops ____ and pulls ____ through faster."]]
5. [[Major step.]]
6. [[Major step.]]
 - [[The failure mode stated as a consequence: "If it goes over the edge, ____ escapes into your ____ and your ____ is wrong."]]
7. [[Rinse-and-transfer step so nothing is left behind.]]
8. Go start Part D while this [[drains / runs / cools]].
9. [[What happens to the product at the end of the period and when it gets massed.]]

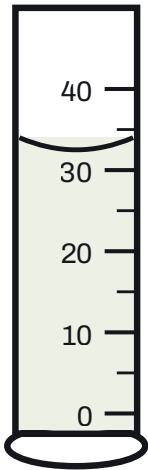
Part D — [[Part Name]] [[DO THIS WHILE PART C RUNS]]

1. [[Identify and describe the sample: color, luster, surface texture.]]
2. [[Measure to a stated precision.]]
3. [[Measure, with the anti-rounding warning: "Record the exact initial volume to 0.1 mL — not '30 mL.'"]]
4. [[Measure.]]
5. [[The subtraction / derived quantity.]]

- 6. [[Calculate. Carry units through the division.]]
- 7. [[Identify from the table below, then calculate percent error against the accepted value.]]
- 8. [[Return the sample to the tray.]]

[[MATERIAL]]	[[SYMBOL]]	[[ACCEPTED VALUE (UNITS)]]
[[name]]	[[sym]]	[[value]]
[[name]]	[[sym]]	[[value]]
[[name]]	[[sym]]	[[value]]
[[name]]	[[sym]]	[[value]]
[[name]]	[[sym]]	[[value]]

[[Conditions the values apply at, and any value that is a range rather than a single number — say why.]]



[[One-line reading instruction for the figure.]]

DATA TABLES

You built these in your pre-lab. Have them open and ready before you touch a [[balance / meter stick / hand lens]]. Fill them in **as you take each measurement** — not at the end of the period from memory.

DISPOSAL AND CLEANUP — LAST FIVE MINUTES

- 1. [[Liquid waste: exact destination and the reason it is permitted. Never imply — state it. "Our state permits acids and bases of 6.0 M and below down the drain, and this filtrate is dilute salt water — nowhere near that limit." VERIFY before printing.]]
- 2. [[Solid waste / product: where it goes, and when it is finally thrown out.]]
- 3. [[Reusable specimen or sample: how it is returned and in what condition.]]
- 4. **Wash all glassware with the Alconox solution in the blue spray bottle.** Spray inside, scrub with a brush, and rinse thoroughly — soap left in a beaker changes the next measurement made in it.
- 5. Rinse with distilled water and invert everything on the rack to dry.
- 6. Wipe your bench, push in your stool, and wash your hands. **Goggles come off last**, once the whole room is finished — not once you are.

POST-LAB ANALYSIS

Answer these in your lab notebook, numbered 1 through [[N]]. Show every calculation with units — an answer with no work and no units earns partial credit at best.

- 1** [[WHAT HAPPENED — two measurements disagreed. State which is closer to true and explain what about the two tools causes the difference.]] **PART B**
- 2** [[CALCULATION — show the calculation, name your result, and calculate percent error against the accepted value. Units in every step.]] **PART D**
- 3** [[WHY — using your recorded observations, explain where the product came from. Refer to the equation in the Background.]] **PART C**
- 4** [[ERROR ANALYSIS — a specific procedural slip. State whether it raises or lowers the calculated value, and say specifically whether the error entered through the ____ or the ____ .]] **ERROR ANALYSIS**
- 5** [[ERROR ANALYSIS — why a step had to happen in the order it did. What would you be unable to determine if you skipped it?]] **ERROR ANALYSIS**
- 6** [[CONNECTION — pick one measurement you made today and identify the correct equipment for it from Section [[0.1]], then name the one safety rule from Section [[0.2]] that applied while you took it.]] **[[U0 / S0.1 · S0.2]]**